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GenAI for Synthetic Data Creation using StyleGAN2-ADA

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Abstract - Deeplearning Models requires large and balanced dataset but medical datasets are limited due to privacy issues, high costs and class imbalance. Conditional GANs(CGANS) and pixel-based mixup methods improves the dataset size but this kind of methods produces unrealistic images that leads to unstable training and this doesnot make them very much useful in the medical applications. This study proposes a better data augmentation and detection framework. It leverages a label-aware generative mixing approach based on StyleGAN2-ADA.In this process we train the generator using the dataset and the dataset is conditioned on the soft labels that leads to generate realistic images.The synthetic images generated will get combined with real data to train the classification model. We measured accuracy, precision, recall, F1 score, macro-F2 and false negative rate. The results are observed that this StyleGAN2-ADA augmentation improves detection performance when compared with the CGAN-based and traditional methods

Keywords: Deep Learning, Data Augmentation, Synthetic Images, CGAN's, Medical Images, StyleGAN2-ADA.

1. INTRODUCTION

Deep learning models have made tremendous success in the recent years in different fields, mainly due to the availability of large and good datasets. Yet, in numerous practical applications particularly in sensitive domains healthcare and medical imaging attaining sufficient data is still a major hurdle. The effectiveness of machine-learning model trained over real data fails due to data privacy issues, a high cost of acquisition and labeling, and insufficiency of rare or critical cases.

Synthetic data generation has come up as a feasible solution to these problems. Researchers can expand datasets with data that closely mimics reality without leaking privacy via the creation of artificial data.

Generative Adversarial Networks(GAN's) belongs to class of generative models which consists of strong ability to learn complex data distributions and generate realistic images. A style-based GAN architecture called StyleGAN2 is widely known for producing the realistic and high resolution images. This new or creative unique design allows a good control over image features. This also guarantee a more stable training process when compared to earlier GAN's These unique characteristics ofStyleGAN2 make it widely useful for various essential applications

Using StyleGAN2 Generative AI for Synthetic Data Generation is examined in this project, which reflects the properties of real datasets. The model learns meaningful patterns and structures that are important. The allows the system create various realistic samples. That are widely useful.The existing dataset can be added to using these samples to rectify the problem of class imbalance in downstream performances.

Altogether this work signify how creating this synthetic data using StleGAN2 can be more useful, effective, scalable, and privacy-friendly choice along with gathering the real data. This helps us in creating strong AI systems this also helps in handling and reducing the real issues of data scarcity.

2. LITERATURE SUREVY

Geometric and Photometric Transformations: This includes a number of standard (and strong!) overfitting reduction approaches including rotations, crops, intensity augmentations. Nevertheless, these approaches do not produce new anatomical configurations or unusual pathologies. [1]

Region-Removal Techniques: Methods such as Cutout black out random patches of an image to push our model to focus on the global context. [2]

CutMix: This takes this further by mixing patches from one image into another, with the labels weighted according to area. [3]

Compositional Techniques: AugMix helps in getting better calibration and robustness by using random chains of simple augmentations and mixing it with the original image. [4]

Standard Mixup: This well-known mixup technique generates examples that are the linear combination of pairs of images in the training process, along with their labels. [5]

Clinical Inconsistency: Most medical images appear to have pixel-wise blending as such augmentations lead to visually unrealistic or clinically implausible samples. [6]

Semantic Ambiguity: Interpolating between classes that are not similar semantically yields semantically ambiguous output which is confusing to a model, or provides incorrect training scalar in safety-critical domains. [7]

Manifold Mixup: This variant generalizes interpolation to hidden feature spaces instead of raw pixels, although generation of realistic anatomical images is still not targeted. [8]

3. PROPOSED SYSTEM

There are two steps in the proposed system. Initially, an unsupervised training is done for a generator to model the data surface. Next step, train the conditional GAN. The learning of the base distribution of medical images is the primary objective at this point. The generators typically use StyleGAN2-ADA architecture which caters to their best performance with less data. This model maps a noise vector and class label to synthetic image. In GAN framework, The Discriminator will act as a binary classifier. It contests against the generator. It the generator. Thus, it learns to discriminate between real CT scans and synthetic fakes.

Input Data: Real images are resized (128 x 128) along with one-hot encodes conditionally supervised. In the second stage of the process, they train the classifier, and then apply augmentation on their generator The model samples a "soft" label vector from a Dirichlet distribution, instead of a standard one hot label. A dominant class is chosen at random. We configure a concentration vector that still has small values for other classes, so as to "bias" the label toward that class. This generator produces an image which is a mixture of these soft labels. So that the resulting image does not become a ghostly overlay as happens in the traditional mixup. The last training set is Real+Generated, which combines real-world data and synthetic data obtained from GANs. Networks such as ResNet-50, ResNet-101 and EfficientNet-B0 are trained on this data.

4. METHODOLOGY

The proposed system leverages a two-stage pipeline that involves first establishing a high-fidelity generative base and then utilizing a novel soft-label interpolation scheme intended to bolster the classification for the medical images.

The system is trained on a dataset named COVIDx-CT-3, consisting of CT scan images from three different classes which are COVID-19, Pneumonia and Normal. All these images are cropped and resized as 128×128 datasets for training the generative model.

4.1 Conditional generative modelling

We use the architecture of StyleGAN2-ADA [31], designed to work well in limited data regimes, for modeling conditional distribution of CT scans.

4.1.1 Installation of system.

We will denote to the starting point of their generative framework as.

$$S_0 = \{G, D, X\ real\} \quad (1)$$

Where:

- G learns the synthesis of 128 x 128 resolution.
- D is the discriminator network that is trained to recognize a synthetic sample.
- X_{real} is the preprocessed **COVIDx-CT-3** dataset.

4.1.2 Objective Function

The networks are optimized via a supervised mini-max game:

$$\min_G\ \max_D\ E_{x\sim p_data}[\log D(x|y)]\ E_{x\sim p_z}[\log(1-D(G(Z|Y)))] \quad (2)$$

- $E_{x\sim p_data}[\log D(x|y)]$: The Discriminator seeks to maximize its ability of distinguishing between real vs generated images.
- $E_{z\sim p_z}[\log(1-D(G(Z|Y)))]$: The Generator (G) tries to minimize the chance that the Discriminator (D) detects its fake images.
- y Represents Condition One-Hot Encoded Label (for example [1,0,0] = COVID)

4.2 Generative Interpolation

We will first introduce a concentration vector that allows to control the "intensivity" of the mix. This vector indicates how much the resulting image will like the dominant class and how much it will like all the others.

$$c \sim \text{Uniform}(\{1,\ldots,K\}) \quad (3)$$

4.2.1 Choices of the Dominant Class

The system picks one of the K classes (COVID, Pneumonia, or Normal) at random to be the "primary" focus of the new image

4.2.2 Construction of Concentration Vector(ϑ)

To tune the "intensity" of the mix, we define a concentration vector. The string of values in this vector controls the bias of how the created image will be biased towards the target class vs other potential classes.

$$\Theta_j = \begin{cases} \alpha_dom & if\ j = c \\ \alpha_oth & otherwise \end{cases} \quad (4)$$

- α_dom=2, α_oth =1: "mix" is controlled by these two parameters.
- To accomplish this, set higher α_dom . It makes it more likely that our synthetic image would be an embodiment of the dominant class but smoothly incorporates some of the other classes features as well.

4.2.3 Dirichlet Soft-Label Sampling (*l*)

Instead of hard-labeling we draw a soft-label vector from a Dirichlet distribution based on the previously established concentration vector:

$$l \sim \text{Dirichlet}(\vartheta) \quad (5)$$

- This samples a certain soft-label vector where (for example $l = [0.7, 0.2, 0.1]$).
- The concept of the diseases is blended together instead of the original pixel values as done in standard mixup.

4.2.4 Synthetic Image Synthesis (x_{syn})

$$X_{syn} = G(z, l) \quad (6)$$

- Generator(G) takes the noise(z) and the soft-label(l) as an input to generate image
- The result X_{syn} is an image that portrays features of the dominating class while containing subtle features of the other classes as defined in the soft-label.

4.3 Classifier Training and Evaluation

Then, real images and generated images are aggregated to train the diagnostic models.

4.3.1 Dataset Stacking

$$D_{augmented} = D_{real} \cup \{(x_i, l_i)\}_{i=1}^{N} \quad (7)$$

- This equation simply indicates that we are augmenting our original real dataset with N synthetic pairs.

4.3.2 Evaluation

$$\text{Macro -F1} = \frac{1}{K}\sum_{i=0}^{k} F1_I \quad (8)$$

- This calculates mean F1-score over all labels.
- The goal is to enlarge this number maximizing the True Positive Rate and minimizing False Negative Rate for COVID-19.

Where e precision and recall of $F1_i$ score are the harmonic mean for a certain class.

$$F1 = 2 \times \frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} \quad (9)$$

- **Precision:** Measures the accuracy of positive predictions.

$$\text{Precision} = \frac{TP}{TP+FP} \quad (10)$$

- **Recall (Sensitivity):** Measures the model ability of recollecting all the positive actual cases.

$$\text{Recall} = \frac{TP}{TP+FN} \quad (11)$$

5.ARCHITECTURE

This depicts the overall framework of the suggested architecture for medical image classification. It shows the process from image generation of GAN training the classifier on this dataset

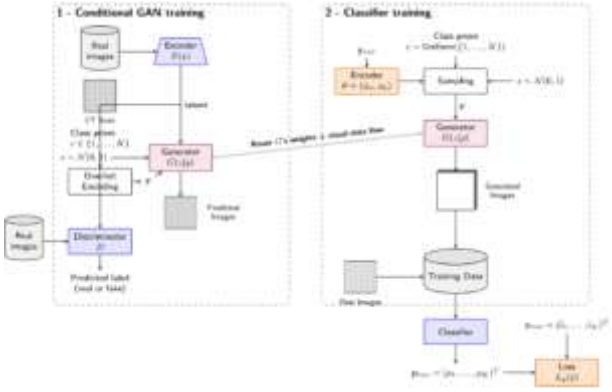

Figure 1: shows Stage 1: A conditional GAN is trained on real images, with one-hot encoded label Stage2 : where the trained generator will generate images using soft labels sampled from a Dirichlet distribution

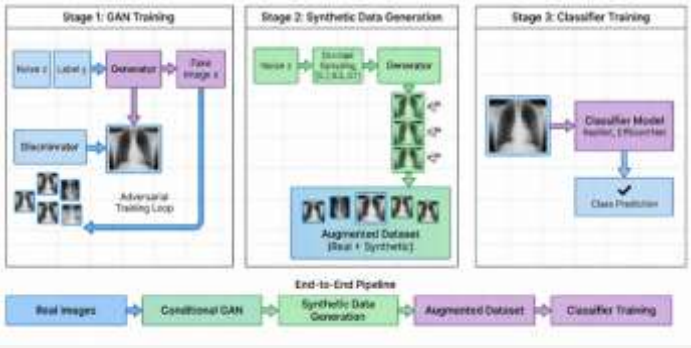

Figure 2: shows a three-stage process for GAN-based medical image augmentation.

6.RESULTS AND DISCUSSIONS

The classification model was enhanced after implementing the proposed technique of generating and augmenting synthetic data as per the experiments.

6.1 Synthetic Data Generation

This depicts that proposed generative framework can yield synthetic images to enhance the training dataset and the data scarcity

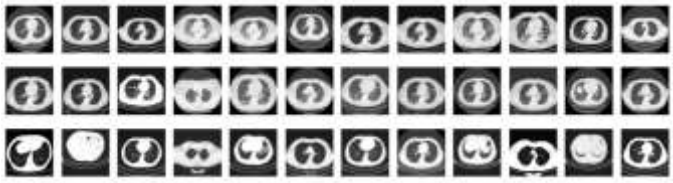

Figure 3 : Shows the generated synthetic images

6.2 Performance Comparison with Baseline Model

The new model does better than the baseline model in terms of its predictive ability.

Metric	Baseline	Our Model
Accuracy	82.4%	91.2%
F1-Score	0.79	0.89
Precision	0.81	0.88
Recall (Sensitivity)	0.76	0.93

Table 1: shows a comparison of performance metrics between the baseline model and our proposed model.

6.3 Statistical Performance

The proposed model is shown to be effective from the result

Statistical performance of the proposed model suggests strong and reliable classification capability. As also shown in the Table 2, the model reaches an accuracy of 91% along with very high macro-averaged evaluation metrics like F1-score, precision and recall indicating a good balanced behaviour across all classes.

Metric	Value
Accuracy	91%
F1-Score(Macro)	0.9075
Precision (Macro)	0.9067
Recall (Macro)	0.9084
Macro-F2 Score	0.9080
False Negative Rate	0.0916

Table 2 : summarizes the overall statistical performance

6.4 Confusion Matrix Analysis

This shows how the model is able to discriminate correctly among the three diagnosis categories indicating an efficacy of the proposed approach

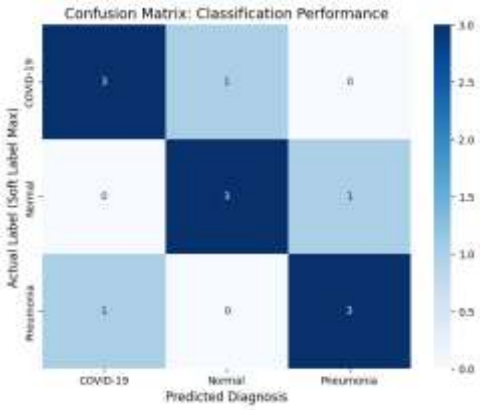

Figure 3 : shows the model's ability to distinguish the three diagnostic categories

6.5 Overall Performance

For the image classification results, we found that our proposed GAN based augmentation method could significantly outperform the baseline model (which was only trained on real life images). Using Dirichlet soft labels we were able to generate synthetic images which enabled the single classifier to learn smoother decision boundaries across the three classes COVID-19, Pneumonia, and Normal.

7. CONCLUSION

This paper proposed the unique generative data augmentation framework, which was an upgrade of the mixup methodology.The heuristic mixing approach interpolates image pairs at pixel-level which limits the semantic compatibility limits of the configuration strongly.Our method however, learns how to mix images and labels by conditioning a StyleGAN2-ADA generator on Dirichlet-sampled soft labels.In doing so, our method generates high-fidelity CT slices semantically aligned to its labels. Experiments have been done on COVIDx-CT-3 dataset and it was found that this model outperformed on ResNet-50 and other architectures.In particular, the macro-F1 scores achieved are better than the baseline and also a significant reduction in false negative rate for COVID-19 detection which a major metric in medical diagnostics.The results confirm that generative mixing with labels can add a much stronger regularization than pixel-blending methods, especially in structural domains. In the future, we will be exploring the generalizability of our system for different applications, natural photobenchmarks, and diverse class imbalance.

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